

HAT-P-32b

R-0238

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_251222 · created 2026-07-17T04:38:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461031.7165 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.014
Transit depth (Rp/Rs)^2	2.0 +/- 0.41 [%]
Orbital Inclination (inc)	89.52 +/- 1.44
Airmass coefficient 1 (a1)	1.1742 +/- 0.0078
Airmass coefficient 2 (a2)	-0.108 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.66 %
Best Comparison Star	None
Optimal Aperture	4.09
Optimal Annulus	7.54
Transit Duration (day)	0.1097 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.014 vs 0.1489 (deviation 0.56σ)
Duration fit vs published	0.1097 d vs 0.1304 d (deviation 9.42σ)
Mid-time O–C	-8.0 min
Depth SNR	10.1
Residual scatter	0.66 %

Light curve

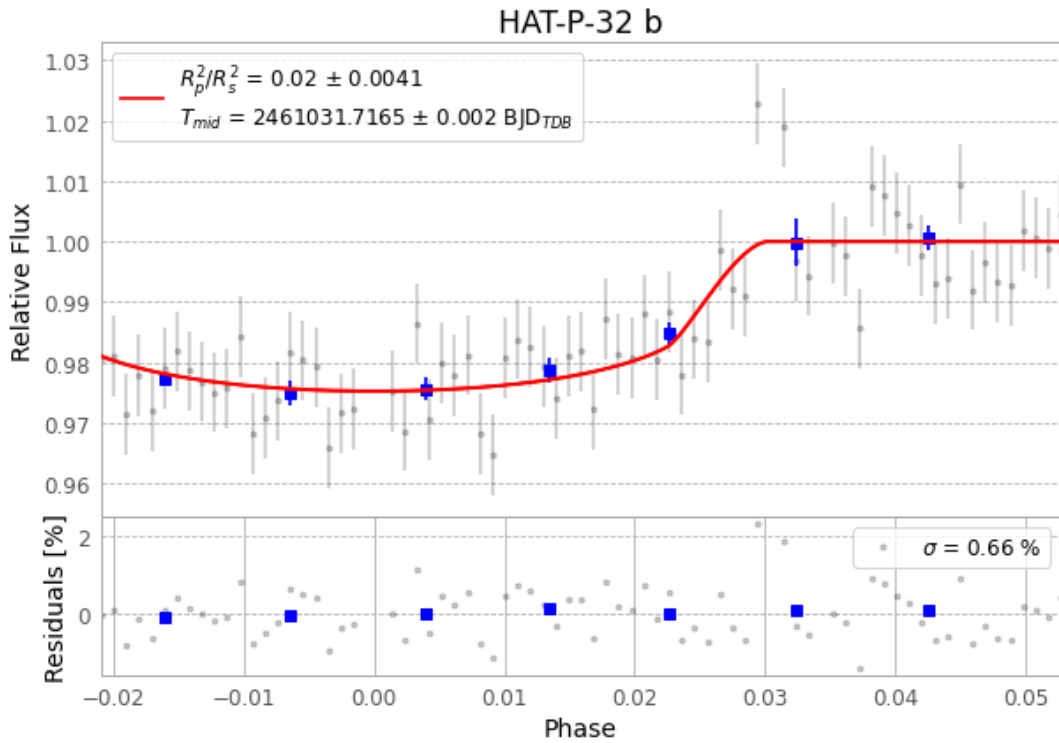


Figure 1 — FinalLightCurve_HAT-P-32 b_2025-12-21.png (SHA-256 4931d1db705708b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 331], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df68a3c192ea6e54...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

77 FITS frames (51 MB), HATP-32251222040015.FITS → HATP-32251222074815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-251222.log (6d39b2632fdc...), FinalLightCurve_HAT-P-32 b_2025-12-21.png (4931d1db7057...), FinalLightCurve_HAT-P-32 b_2025-12-21.csv (248fecb3ef6a...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 04:38Z.